Experiment.1

FTK Imager: A Forensic Imaging Tool Overview

Acquiring Volatile Memory (RAM) Using FTK Imager

Link for installation file :- link

Steps to Capture RAM Using FTK Imager

1. Run as Administrator

- Open FTK Imager with administrative privileges.
- Right-click the FTK Imager icon and select Run as administrator
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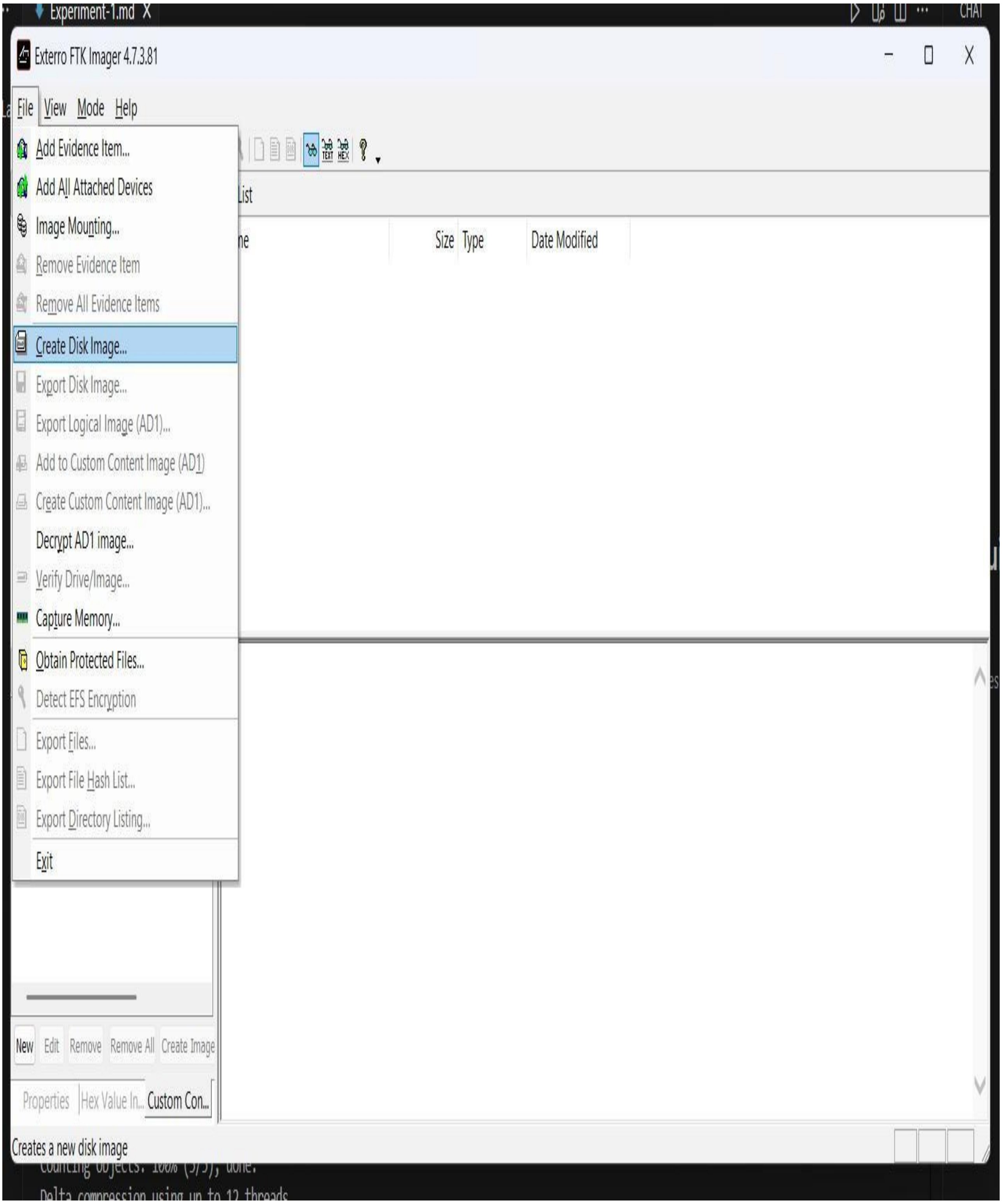
2. Initiate Memory Capture

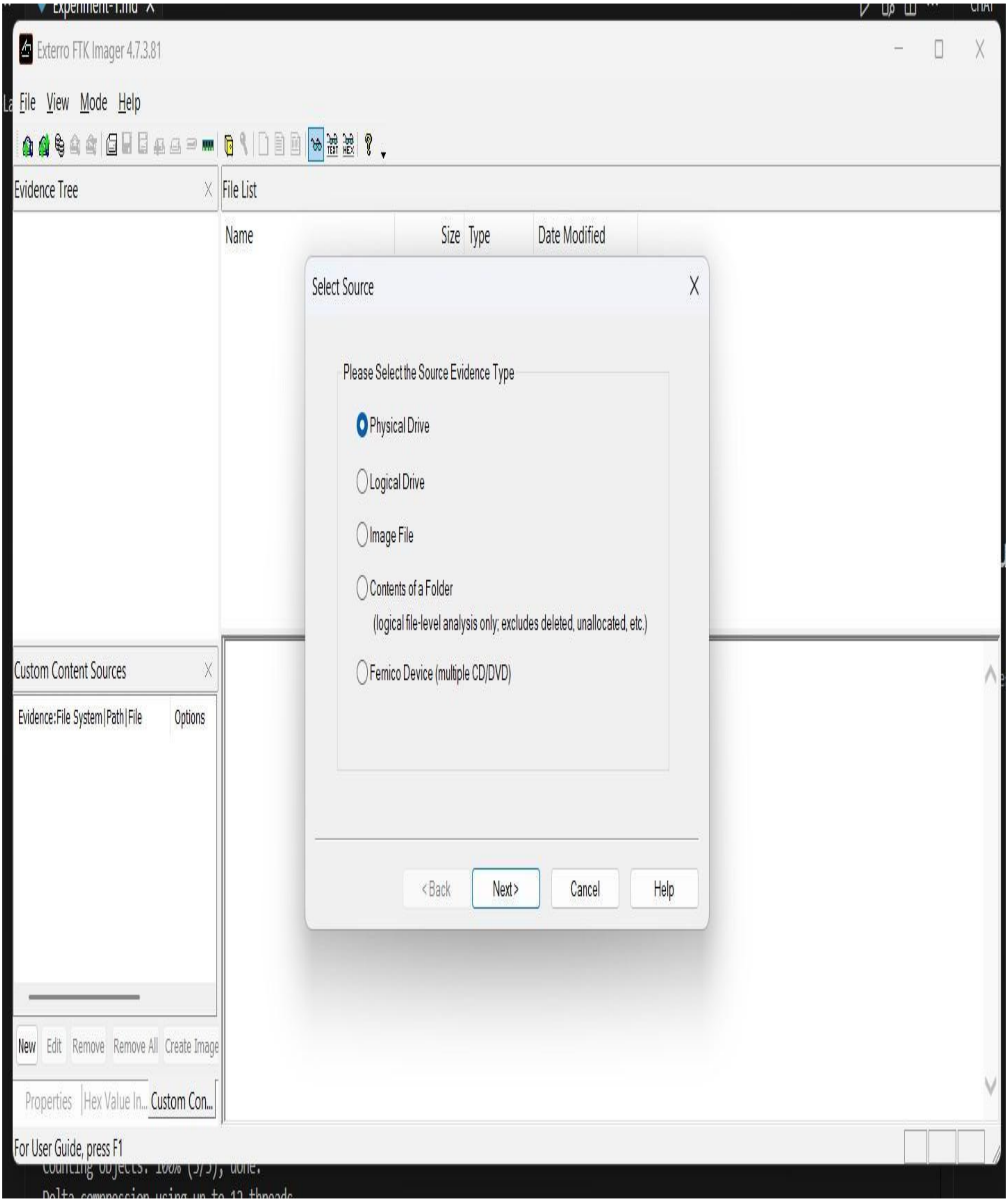
- In the top menu bar, click File → Capture Memory from the dropdown list.



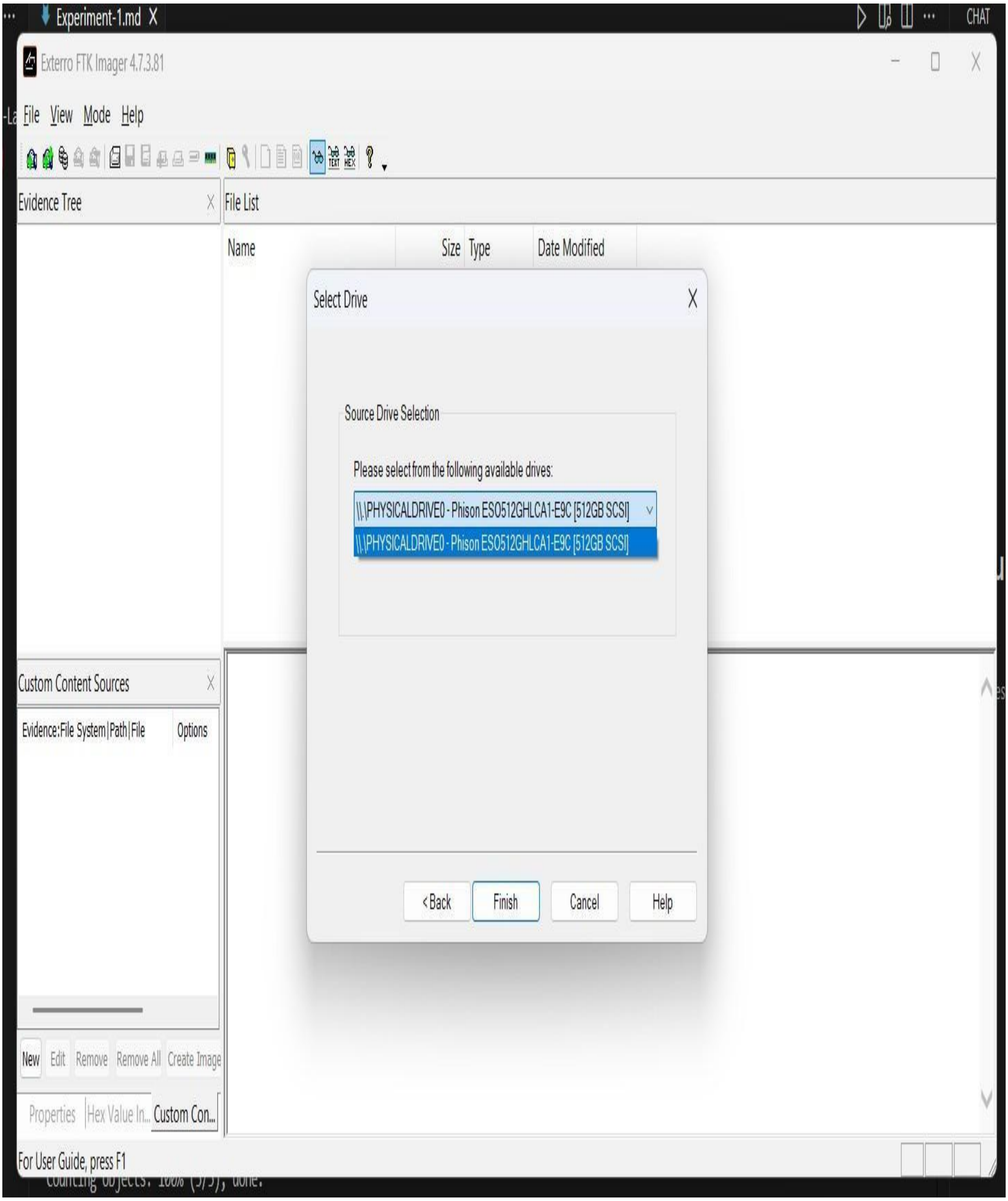
3. Configure Destination

A dialog box will appear where you configure where and how the memory will be saved.

- **Destination Path:**
  Click **Browse** to select a folder on an **external drive** (not the system drive).
- **Destination Filename:**
  You can keep the default memdump.mem or assign a more descriptive name.

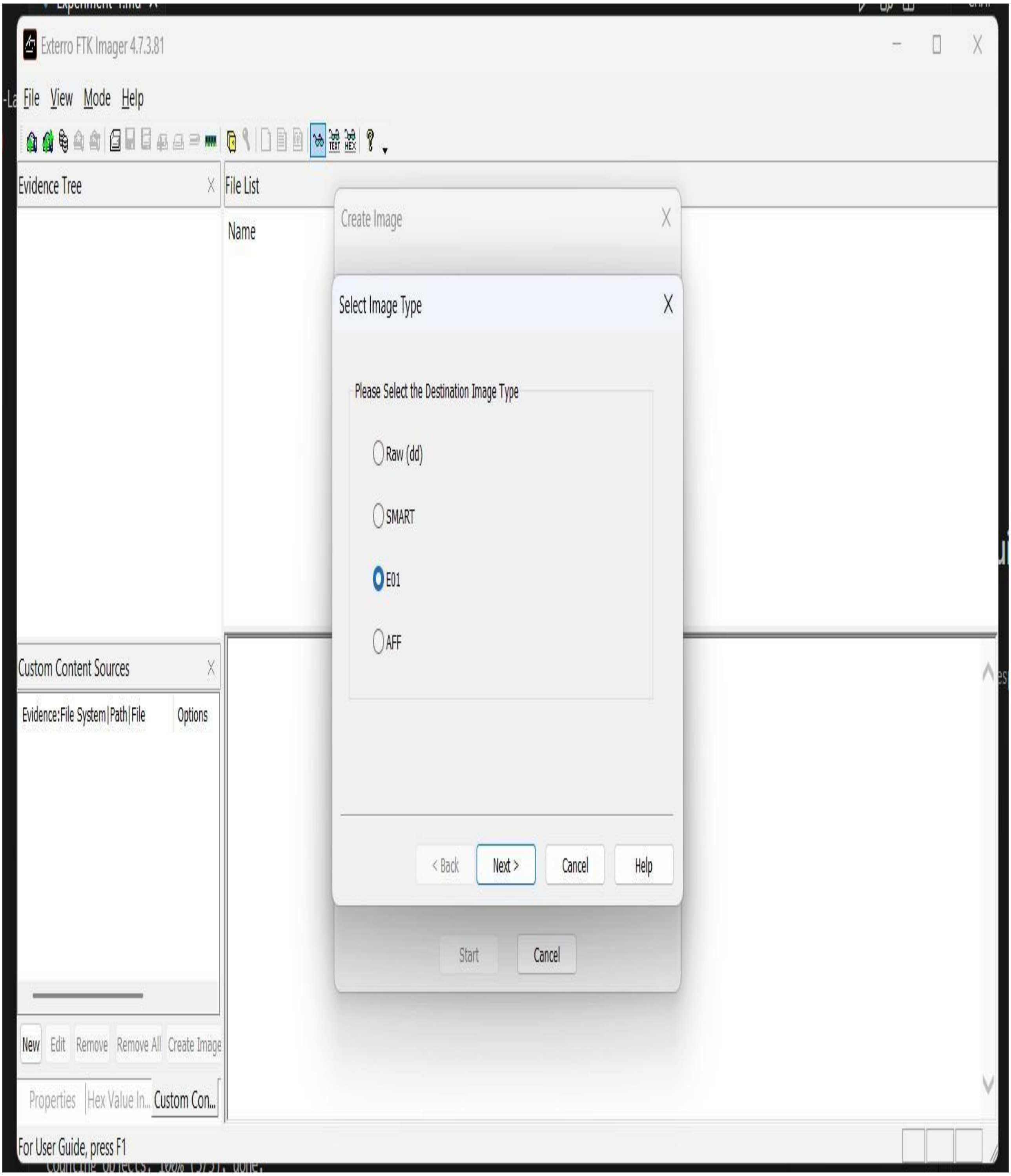


- **Optional: Include pagefile.sys**
  Check this box to capture pagefile.sys, which is virtual memory stored on the disk.
- **Optional: Create AD1 file**
  Saves the memory dump in an AccessData-specific container file.



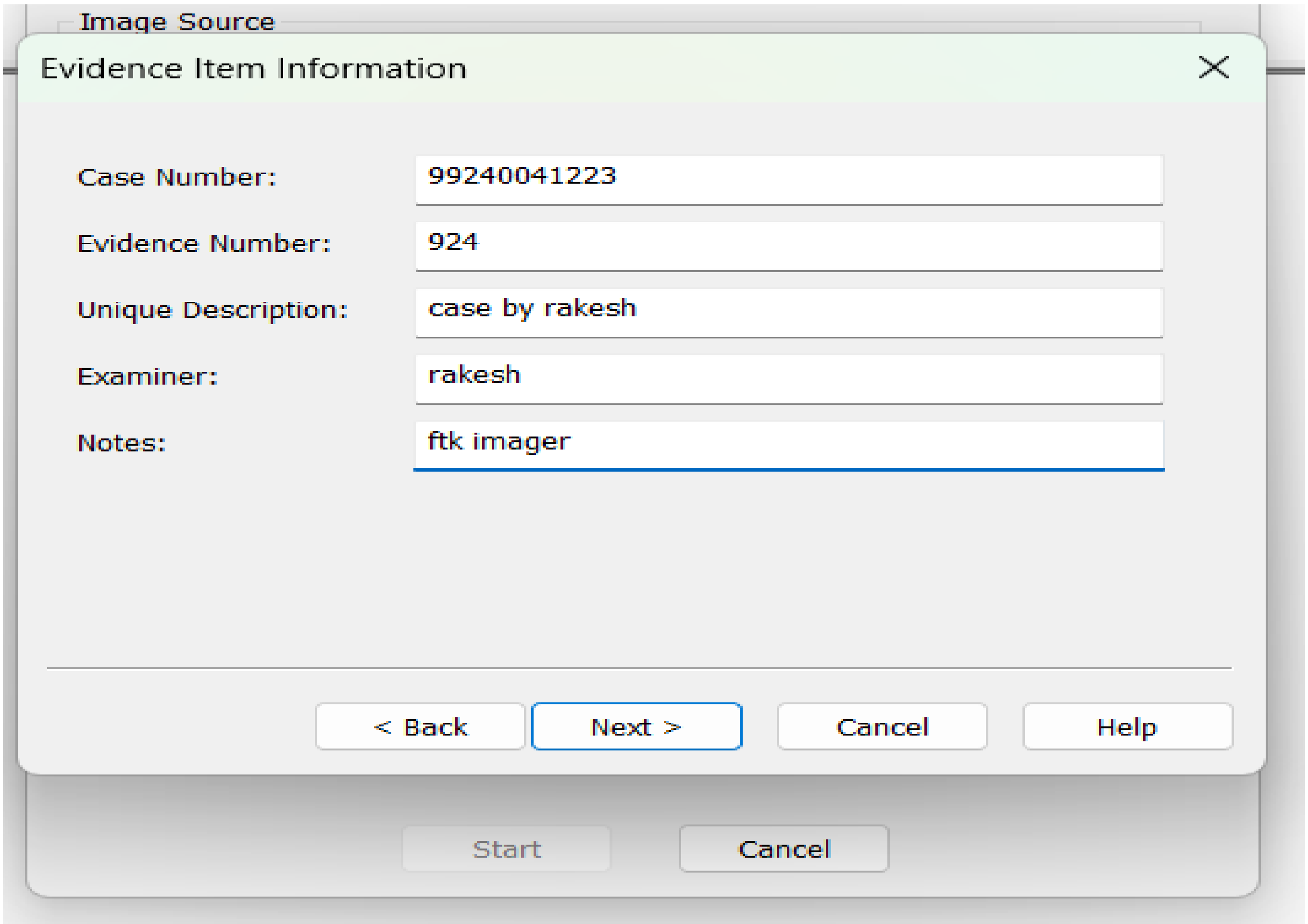
4. Start Capture

- Click the **Capture Memory** button to begin acquisition.

- 

5. Wait for Completion

- A progress bar will indicate the capture status.
- Capture time depends on the system's RAM size.
- Once finished, the memory dump file will be available in the destination folder.



Acquiring Non-Volatile Memory (Disk Image) Using FTK Imager

1. Start the Process

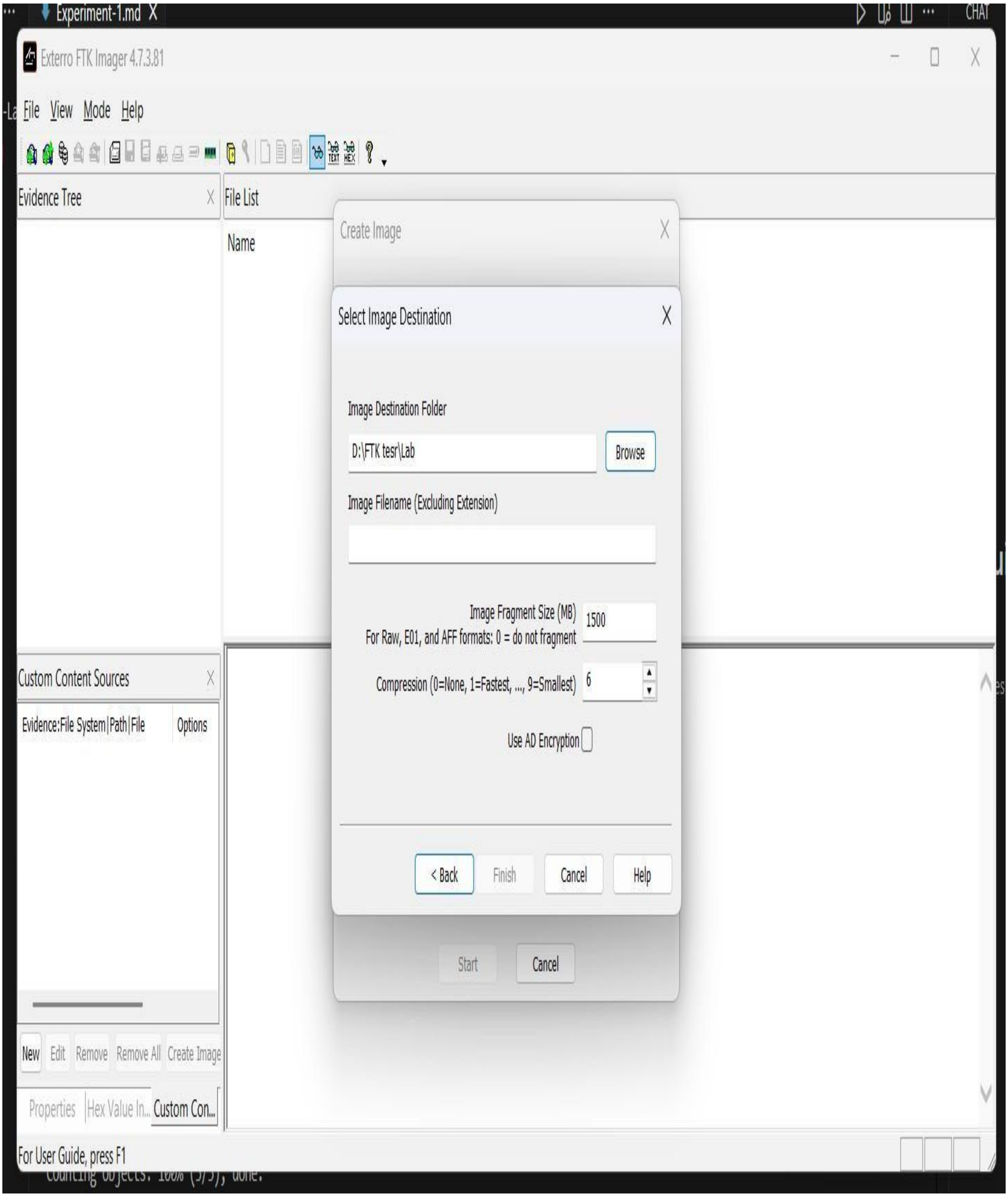
- In FTK Imager, go to the top menu bar: File → Create Disk Image....

2. Select Source Evidence Type

A window will appear asking you to choose the source type:

- **Physical Drive:** Images the entire disk, including all partitions, unallocated space, and the Master Boot Record (MBR).

- **Logical Drive:** Images a specific partition (e.g., C: drive).
- **Image File:** Converts or copies an existing image file.
- **Contents of a Folder:** Creates an image of a specific folder only.

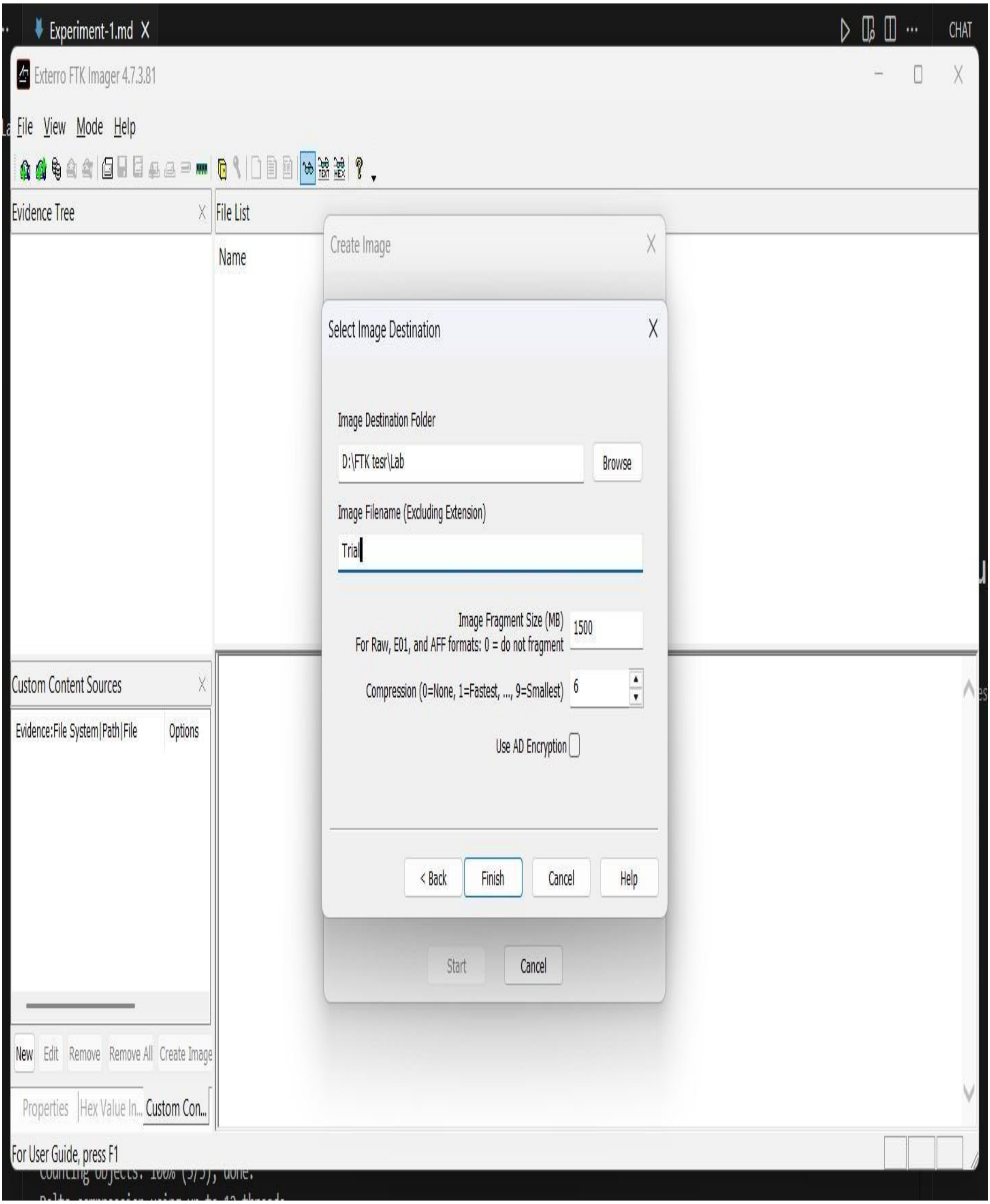


3. Select the Source Drive

- From the dropdown, choose the physical drive to image (connected via **write-blocker**).
- Click **Finish**.

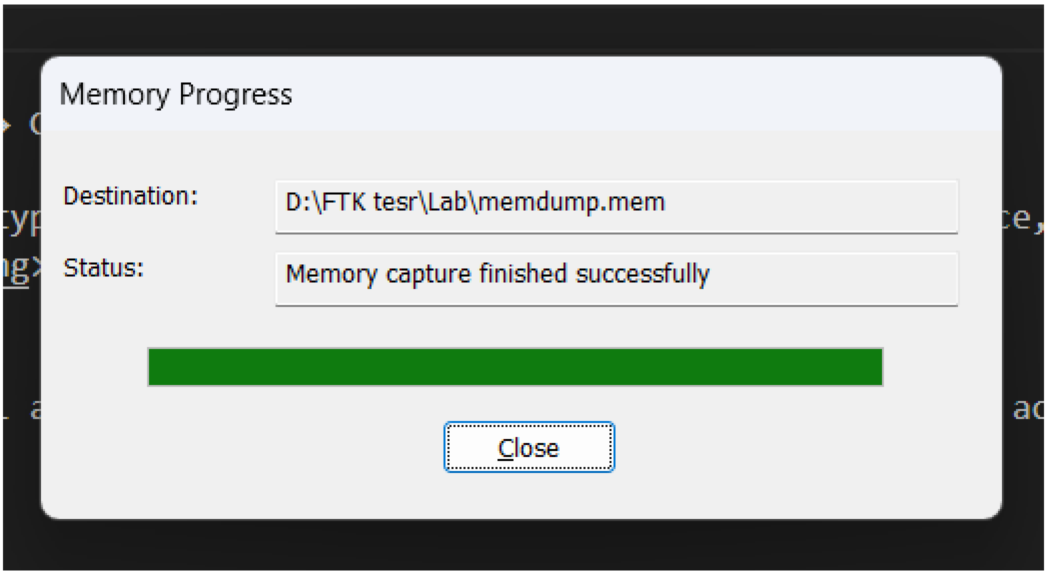
4. Configure the Image Destination

- Click **Add...** in the "Create Image" window to define the image **format** and **destination**.



Options:

- **Image Type:**
  - **E01 (EnCase Format):** Recommended; includes compression, metadata, and error-checking.
  - **Raw (DD):** Bit-for-bit copy with no extra features.



- **Fill in Evidence Item Information:**
    - Enter case details, examiner name, and description.
    - This information is stored in the image metadata (important for documentation).

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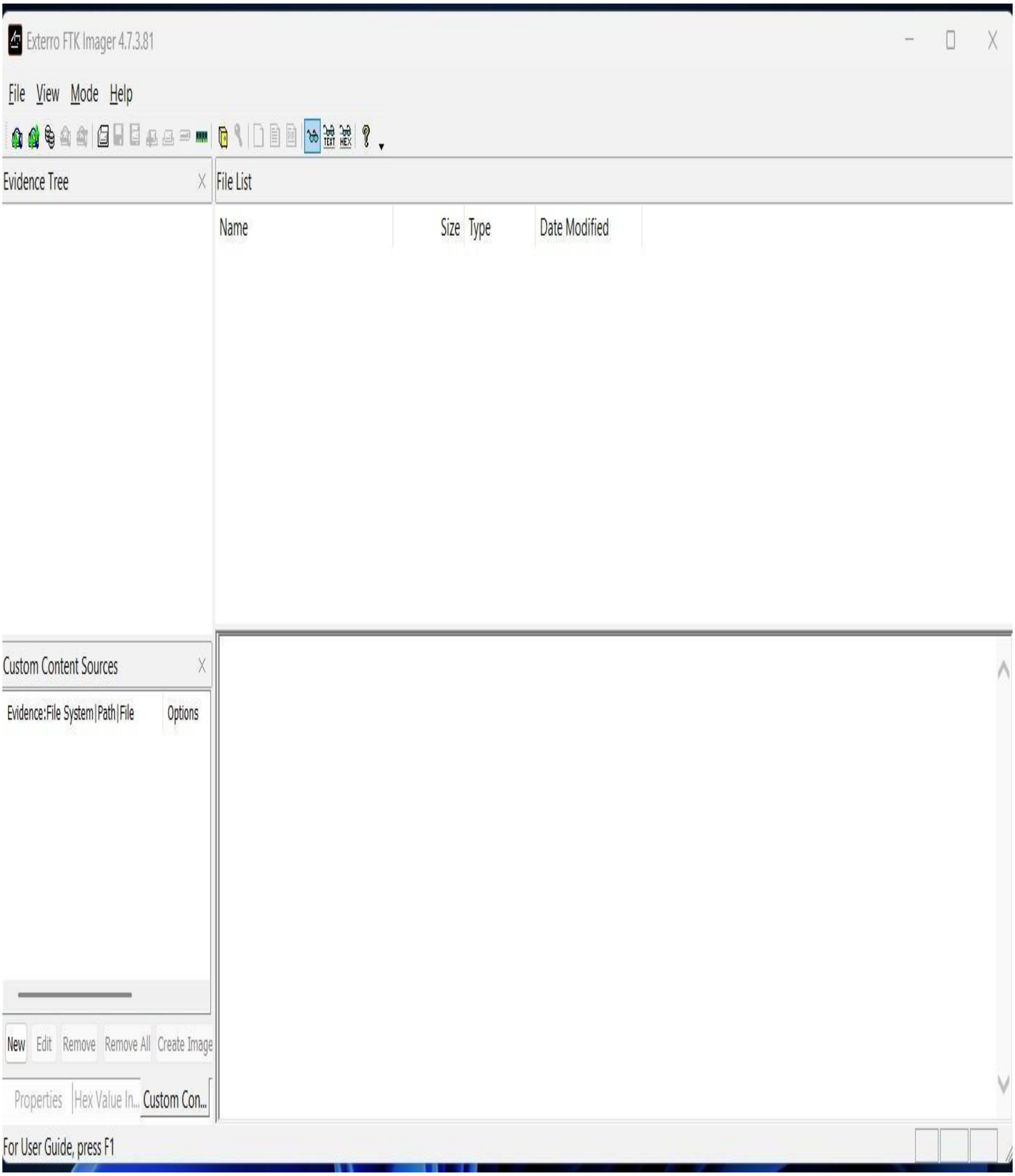
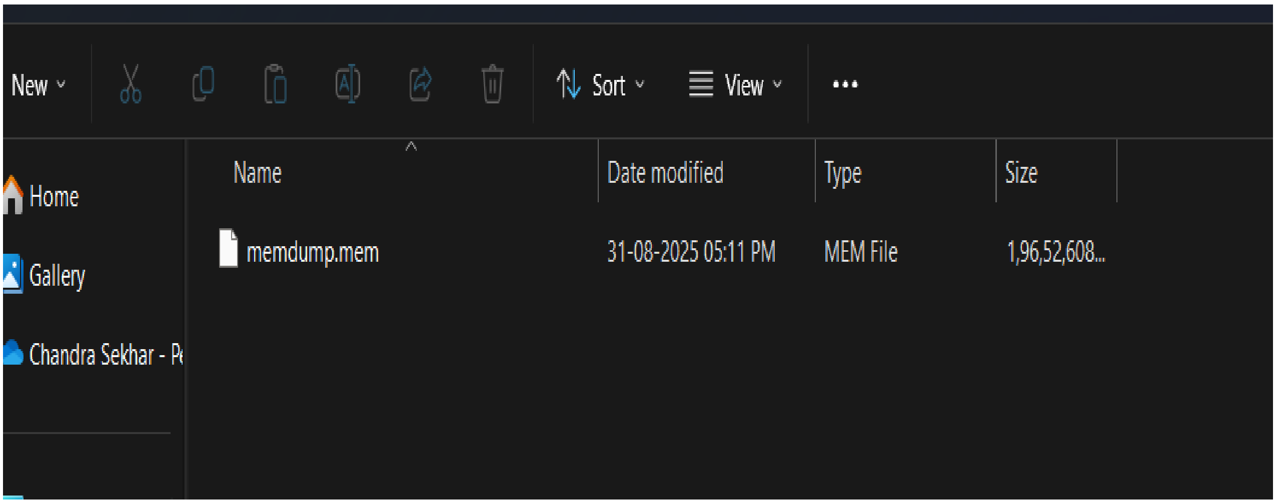
- **Choose Destination Folder:**
    - Must be a different drive from the source.
    - Name the image file (e.g., Case001_SuspectHDD).
- **Image Fragment Size:**
    - Set a value to split the image into multiple parts.
    - Set to 0 for a single image file.

5. Start the Imaging Process

- Click **Finish** to return to the "Create Image" screen.
- **Check "Verify images after they are created"** to calculate hash values and ensure integrity.
- Click **Start**.

6. Completion and Hash Verification

- The imaging process may take time depending on the drive size.
- After completion, FTK Imager verifies the hashes automatically.
- A final window shows **MD5** and **SHA1** hashes for both the source drive and image.
- Matching hashes confirm the forensic image's integrity.

- 
Exterro FTK Imager 4.7.3.81
File View Mode Help
Evidence Tree
File List
Name
Size
Type
Date Modified
Custom Content Sources
Evidence:File System|Path|File
Options
New
Edit
Remove
Remove All
Create Image
Properties
Hex Value In...
Custom Con...
For User Guide, press F1
- 
New
Sort
View
Home
Gallery
Chandra Sekhar - P
Name
Date modified
Type
Size
memdump.mem
31-08-2025 05:11 PM
MEM File
1,96,52,608...

References

- FTK Imager Official Website
- FTK Imager Documentation